

Protocol for data collection on drugs use in inpatients

Forms used for data collection in the study entitled: "Risk factors for adverse drug reactions in pediatric inpatients: a cohort study"

Paulo Henrique Santos Andrade^{1*}, Iza Maria Fraga Lobo^{2¶}, Wellington Barros da Silva^{1¶}

¹Universidade Federal de Sergipe, São Cristóvão, Sergipe, Brazil;

²Hospital Universitário de Sergipe, Aracaju, Sergipe, Brazil.

* Corresponding author

E-mail: paulo_henrique0@hotmail.com (PHSA)

¶ These authors also contributed equally to this work

Guidelines

Forms 1 and 2 must be used in the first pharmaceutical consultation after hospital admission of the patient:

1. form 1 is aimed at obtaining information on the prior history of ADR of the patient; and
2. form 2 is aimed at obtaining information on the prior history of use of the drug.

Forms 3-6 must be used for the daily recording of information observed in medical prescriptions.

1. form 3 is used to record potential drug interactions;
2. form 4 for registration of off-label drugs prescribed;
3. form 5 for the daily recording of the use of drugs; and
4. form 6 for daily monitoring of prescription errors.

Furthermore, form 7 must be used for the registration of abnormal laboratory tests or other suspected adverse events.

Form 1: Prior history of ADR of the patient

BRIEF ACCOUNT OF THE OCCURRENCE: ☐ ALLERGY ☐ OTHER ADR

[illegible]

DRUG SENSITIVITY TEST PERFORMED? ☐ YES ☐ NO

MEDICAM RECORD

Are there are signs of the self-report of ADR? ☐ YES ☐ NO

At where? ☐ PRESCRIPTION ☐ MEDICAL RECORDS ☐ HOSPITAL BED

Was there PRESCRIPTION of the self-reported drug to the patient? ☐ YES ☐ NO

(Check with a * the drugs in which this occurred)

Was there ADMINISTRATION of the self-reported drug to the patient? ☐ YES ☐ NO

(Check with a * the drugs in which this occurred)

Form 2: Medication reconciliation on hospital admission

MEDICINES IN USE BEFORE ADMISSION

	NAME OF THE DRUG	DOSE	ROUTE	FREQUENCY	DURATION OF THERAPY
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					

WAS ANY OF THESE MEDICINES PRESCRIBED AFTER ADMISSION? ☐ YES ☐ NO

WAS A PRESCRIBED MEDICATION OF THE SAME MEDICINAL CLASS? ☐ YES ☐ NO

TO WHICH DRUGS? _____, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____.

WAS THERE CHANGE IN DOSE? ☐ YES ☐ NO

WHICH DRUGS? _____, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____.

WAS THERE CHANGE IN ROTE? ☐ YES ☐ NO

WHICH DRUGS? _____, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____.

WAS THERE CHANGE IN FRENQUENCY? ☐ YES ☐ NO

WHICH DRUGS? _____, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____.

WAS THERE CHANGE IN DURATION OF THERAPY? ☐ YES ☐ NO

WHICH DRUGS? _____, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____.

Form 3: Potential drug interactions

DRUG INTERACTIONS	GRAVITY	ADE POTENTIAL	OCCURRENCE
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			
			☐ YES
			☐ NO
Duration in days:			

HOW MANY MEDICINAL INTERACTIONS WERE FOUND?

Form 4: Off-label use

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Inadequate prescription: _____

Proper prescription: _____

Has there been an off-label use event? ☐ YES ☐ NO, IF YES, REPORT: _____

Does the child make use of drugs for continuous use? ☐ YES ☐ NO

Are there SERIOUS drug interactions in the prescription? ☐ YES ☐ NO

Form 6: Monitoring of prescription errors

MONTH/YEAR ____/____

Prescription errors	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	Σ
Nº drugs without a dose																																
Nº drugs without a route of administration																																
Nº drugs without dosage																																
Nº drugs prescribed by trade name																																
Nº prescription drugs with vague expressions																																
Nº prescription drugs in a way that unreadable																																
Nº injectable drugs prescribed																																
Nº drugs without infusion rate																																
Nº drugs without infusion time																																
Nº drugs without diluent																																
Nº drugs without diluent volume																																
Nº prescribed antibiotics																																
Nº antibiotics without treatment duration																																
Nº prescription drugs																																
Nº drugs prescribed with error																																

POP DE PREENCHIMENTO

- ✓ Nº drugs without a dose – the dose is not written or there is no way to calculate it.
- ✓ Nº drugs without a route of administration – absence of route of administration, which must be clearly prescribed.
- ✓ Nº drugs without dosage – absence of dose interval.
- ✓ Nº drugs prescribed by trade name – quantity of drugs described by trade name.
- ✓ Nº prescription drugs with vague expressions – medication without conditions of use. The dosage, maximum daily dose and condition that determines the use or interruption of the use of the drugs should be clearly described.
- ✓ Nº prescription drugs in a way that unreadable: quantity of drugs with illegible description.
- ✓ Nº injectable drugs prescribed – quantity of injectable drugs prescribed for each day according to each patient.
- ✓ Nº drugs without infusion rate – only for intravenous drugs.
- ✓ Nº drugs without infusion time – only for intravenous drugs.
- ✓ Nº drugs without diluent – the intravenous, intramuscular, subcutaneous and neuraxial and nerve plexus drugs, the prescription should contain information on diluent (type and volume).
- ✓ Nº drugs without diluent volume – analyze the amount of solution described for the dilution.
- ✓ Nº prescribed antibiotics – amount of antibiotics prescribed.
- ✓ Nº antibiotics without treatment duration – antibiotics without the amount of days for treatment.

Form 7: Laboratory and clinical accompaniment

Exams		Parameters	/	/	/	/	/	/	/	/	/	Corresponding	
Hemogram	Hemoglobin	12-17 g/dL										Myelotoxicity	☐ Anemia
	Hematocrit	36-57 %											
	Neutrophil	2000-7700/mm³											☐ Neutropenia
	Eosinophils	40-660/mm³											☐ Eosinophilia
	Lymphocytes	800-4400/mm³											☐ Lymphocytopenia
	Monocytes	120-990/mm³											☐ Monocytopenia
	Basophile	0-200/mm³											☐ Basophilia
	Platelets	150-400.10³ K/uL											☐ Thrombocytopenia
Liver function	AST	10-39 (U/L)										Myelotoxicity	☐ Hepatotoxicity
	ALT	10-45 (U/L)											
	ALP	0-645 (U/L)											
	GGT	5-58 (U/L)											
	DB	0,0-0,4 mg/dL											
	IB	0,0-0,8 mg/dL											
	Albumin	3,5-5,5 g/dL											
	Globulin	3,5-5,5 g/dL											
Renal function	Creatinine	0.5-1.2 mg/dL										Myelotoxicity	☐ Nephrotoxicity
	Urea	15-45 mg/dL											
	Uric acid	1,5-7,0 mg/dL											
Electrolytes	Sodium	136-145 mmol/L										Myelotoxicity	☐ Electrolyte disturbances
	Potassium	3,5-5,1 mmol/L											
	Calcium	8,8-11 mg/dl											
	Magnesium	1,58-2,56 mmol/L											
	Chloride	98-110 mmol/L											
	LDH	200,0-480,0 U/L											
	Lactate	4,5-19,8 mg/dL											

Pharmacodermia			Gastrointestinal toxicity			Neurotoxicity			Cardiotoxicity		
Patches on the skin	/	/	/	/	/	Agitation	/	/	Change - ECG	/	/
Pruritus	/	/	/	/	/	Seizure	/	/	Tachycardia	/	/
Hyperemia	/	/	/	/	/	Consciousness change	/	/	Hypotension	/	/
Stevens-Johnson Synd.	/	/	/	/	/	Unintentional movement	/	/	Shock	/	/